Mathematics of Cryptography Tutorial Roadmap

Purpose

This document is a proposed learning roadmap for building the mathematical foundation needed to understand classical and modern cryptography. It is organized in a sensible order, beginning with mathematical notation and bit-level thinking, then moving into modular arithmetic, classical ciphers, finite fields, symmetric cipher design, cryptanalysis, public-key cryptography, elliptic curves, hash functions, and post-quantum cryptography.

The goal is not merely to memorize formulas. The goal is to build mathematical intuition piece by piece, especially for someone learning cryptography through experimentation, visualization, and careful self-study.

Phase 0: Learning the Language of Mathematical Thinking

These tutorials are about becoming comfortable with the grammar of math before diving into cryptography proper.

1. How Mathematicians Use Symbols

Goal: Learn how to read mathematical notation without panic.

Topics:

- Variables
- Expressions
- Equations
- Functions
- Subscripts and superscripts
- Summation notation
- Modular notation
- Set notation
- “For all” and “there exists”

Why it matters:

Cryptography papers and textbooks often look harder than they are because the notation is compressed. This tutorial would be like learning to read a map legend before entering the wilderness.

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2. Functions as Machines

Goal: Understand functions as input-output machines.

Topics:

- Domain and range
- One-to-one functions
- Onto functions
- Composition of functions
- Inverse functions
- Permutations as functions
- Why invertibility matters in encryption

Cryptography connection:

A cipher is usually a function:

$$C = E_K(P)$$

and decryption is the inverse:

$$P = D_K(C)$$

This tutorial connects directly to cipher components in a modular cryptography workbench.

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3. Binary, Hexadecimal, and Bases

Goal: Become fluent in representing values as bits, bytes, hex, and integers.

Topics:

- Base 2
- Base 10
- Base 16
- Converting between binary, decimal, and hex
- Bit positions
- Nibbles and bytes
- Overflow
- Word size

Cryptography connection:

Modern cryptography lives in bits. If binary and hex feel natural, cipher internals become much easier to see.

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4. Bitwise Operations as Algebra

Goal: Understand exclusive OR, AND, OR, NOT, shifts, and rotations mathematically and visually.

Topics:

- Exclusive OR
- Bit masks
- Left shift
- Right shift
- Rotation
- Bit selection
- Bit permutation
- Why exclusive OR is reversible
- Why AND loses information

Cryptography connection:

This is foundational for stream ciphers, block ciphers, one-time pads, Feistel networks, substitution-permutation networks, and key schedules.

Phase 1: Modular Arithmetic and Classical Cipher Math

This is where the cryptography math begins to feel real.

5. Modular Arithmetic: Clock Arithmetic for Cryptography

Goal: Understand wraparound arithmetic.

Topics:

- Remainders
- Congruence
- Addition modulo n
- Subtraction modulo n
- Multiplication modulo n
- Negative numbers modulo n
- Why modulo 26 matters in classical ciphers

Cryptography connection:

Caesar, Vigenère, affine ciphers, RSA, Diffie-Hellman, elliptic curves, and finite fields all use modular thinking.

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6. The Caesar Cipher as Modular Addition

Goal: See one of the simplest ciphers as a mathematical function.

Topics:

- Mapping letters to numbers
- Encryption as addition modulo 26
- Decryption as subtraction modulo 26
- Key space
- Brute force attacks

Example:

$$C = P + K \pmod{26}$$

$$P = C - K \pmod{26}$$

Cryptography connection:

This is the first bridge between letters and algebra.

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7. Multiplicative Inverses Modulo n

Goal: Understand when multiplication can be reversed.

Topics:

- What an inverse means
- Why not every number has an inverse modulo n
- Relatively prime numbers
- Greatest common divisor
- Units modulo n

Example:

$$5^{-1} \pmod{26}$$

Cryptography connection:

The affine cipher, RSA, Diffie-Hellman, and elliptic curve cryptography all rely on modular inverses.

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8. The Euclidean Algorithm

Goal: Learn how to find the greatest common divisor efficiently.

Topics:

- Greatest common divisor
- Repeated division
- Remainders
- Why the algorithm works intuitively

Cryptography connection:

The Euclidean algorithm is one of the workhorses of cryptography. It tells us whether inverses exist.

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9. The Extended Euclidean Algorithm

Goal: Learn how to calculate modular inverses.

Topics:

- Bézout coefficients
- Back-substitution
- Finding $a^{-1} \pmod{n}$
- Why this matters for decryption

Cryptography connection:

This is essential for:

- Affine cipher decryption
- RSA private key generation
- Elliptic curve cryptography
- Finite field arithmetic

This tutorial should be built very gently, because this is one of those places where people often almost get it and then the floor tilts sideways.

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10. The Affine Cipher

Goal: Combine modular multiplication and modular addition.

Topics:

- Encryption formula
- Decryption formula
- Valid and invalid keys
- Why a must be invertible modulo 26
- Breaking affine ciphers

Formula:

$$C = aP + b \pmod{26}$$

Cryptography connection:

This is a beautiful early example of how invertibility determines whether a cipher is usable.

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11. Frequency Analysis and Statistical Leakage

Goal: Understand how patterns survive weak encryption.

Topics:

- Letter frequency
- Monograms
- Digrams
- Trigrams
- Index of coincidence
- Why substitution ciphers leak structure

Cryptography connection:

This is the beginning of cryptanalysis.

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12. The Vigenère Cipher and Repeating-Key Weakness

Goal: Understand polyalphabetic substitution and why repeating keys are dangerous.

Topics:

- Repeated keys
- Modular addition by position
- Kasiski examination
- Index of coincidence
- Key-length detection

Cryptography connection:

This leads naturally into stream ciphers and the importance of never reusing one-time pad keys.

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13. The One-Time Pad

Goal: Understand perfect secrecy in its simplest mathematical form.

Topics:

- Exclusive OR version of the one-time pad
- Modular addition version
- Key randomness
- Key length
- Key reuse disaster
- Perfect secrecy intuition

Formula:

$$C = P \oplus K$$

$$P = C \oplus K$$

Cryptography connection:

This is one of the most important conceptual anchors in all of cryptography.

Phase 2: Permutations, Combinatorics, and Classical Machines

These tutorials connect strongly to rotor machines, transposition, and historical cryptography.

14. Permutations as Rearrangement Functions

Goal: Understand permutations as reversible shuffles.

Topics:

- Permutation notation
- Input position to output position
- Inverse permutations
- Composition of permutations

- Cycle notation

Cryptography connection:

Permutations appear in:

- Transposition ciphers
- Rotor machines
- Substitution-permutation networks
- Block cipher diffusion layers

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15. Counting Keys and Keyspaces

Goal: Learn how to count possible cipher configurations.

Topics:

- Multiplication rule
- Factorials
- Combinations
- Permutations
- Keyspace size
- Brute force feasibility

Cryptography connection:

This helps explain why some ciphers are easy to brute force and others are not.

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16. Transposition Ciphers

Goal: Understand encryption by rearranging positions rather than changing symbols.

Topics:

- Columnar transposition
- Rail fence cipher
- Route ciphers
- Permutation keys
- Inverse permutation for decryption

Cryptography connection:

This prepares the mind for permutation layers in modern block ciphers.

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17. Rotor Machines as Composed Permutations

Goal: Understand machines like Enigma as chains of permutations.

Topics:

- Rotor wiring
- Stepping
- Reflector
- Plugboard
- Composition
- Inverse paths
- Why Enigma encryption is reciprocal

Cryptography connection:

This connects classical mechanical cryptography to modern function composition.

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18. Polyalphabetic Ciphers as Changing Functions

Goal: Understand ciphers where the substitution changes over time.

Topics:

- Multiple alphabets
- Rotor position as state
- Keystreams
- State transitions
- Periodicity

Cryptography connection:

This bridges Vigenère, rotor machines, and stream ciphers.

Phase 3: Probability, Randomness, and Information

This phase is essential for understanding why a cipher looks random and why that is not always enough.

19. Basic Probability for Cryptography

Goal: Understand uncertainty numerically.

Topics:

- Probability as ratio
- Independent events
- Dependent events
- Conditional probability
- Uniform distributions
- Biased distributions

Cryptography connection:

Security often means an attacker cannot distinguish one possibility from another better than chance.

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20. Random Variables and Distributions

Goal: Understand outputs as distributions, not just individual values.

Topics:

- Random variables
- Uniform distribution
- Biased distribution
- Expected value
- Variance, gently introduced
- Sampling experiments

Cryptography connection:

Useful for entropy, randomness testing, avalanche testing, and cryptanalysis.

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21. Entropy as Uncertainty

Goal: Understand entropy in an intuitive cryptographic sense.

Topics:

- Surprise
- Uncertainty
- Shannon entropy
- Min-entropy
- Entropy of keys
- Entropy of plaintext
- Entropy of ciphertext

Cryptography connection:

Entropy helps describe key strength, randomness quality, and information leakage.

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22. Perfect Secrecy and Shannon’s Model

Goal: Understand what it means for ciphertext to reveal no information.

Topics:

- Plaintext space
- Ciphertext space
- Key space
- Conditional probability
- Shannon’s perfect secrecy
- Why one-time pads work

Cryptography connection:

This is one of the philosophical foundations of cryptography.

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23. Randomness Testing and the Limits of “Looks Random”

Goal: Learn why passing visual randomness tests does not prove security.

Topics:

- Byte distribution
- Chi-square test
- Runs test
- Avalanche tests
- Correlation
- Randomness versus unpredictability

Cryptography connection:

Very relevant to experimental cipher work.

Phase 4: Boolean Algebra and Logic

This is important for modern symmetric cryptography, S-boxes, and bit-level design.

24. Boolean Algebra for Cryptography

Goal: Understand logic operations as algebraic structures.

Topics:

- Boolean variables
- Truth tables
- AND, OR, NOT, exclusive OR
- Expressions
- Simplification
- Algebraic normal form

Cryptography connection:

Modern ciphers are built out of Boolean functions.

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25. Exclusive OR as Addition Modulo 2

Goal: Connect bit operations to algebra.

Topics:

- Bits as values in $GF(2)$
- Exclusive OR as addition
- AND as multiplication
- No carries
- Cancellation

Cryptography connection:

This is the stepping stone into finite fields and S-box analysis.

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26. Boolean Functions and Truth Tables

Goal: Understand how a collection of input bits maps to output bits.

Topics:

- n-bit input functions
- One-output Boolean functions
- Multi-output Boolean functions
- Truth tables
- Balanced functions

- Nonlinear functions

Cryptography connection:

S-boxes are vector Boolean functions.

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27. Algebraic Normal Form

Goal: Express Boolean functions as polynomial-like formulas.

Topics:

- Exclusive OR sums
- AND products
- Degree of a Boolean function
- Linear terms
- Nonlinear terms

Cryptography connection:

Useful for understanding algebraic attacks and S-box strength.

Phase 5: Finite Fields

This is where the $GF(2^8)$ tutorial belongs.

28. From $GF(2)$ to $GF(2^n)$

Goal: Understand why finite fields can be built from bits.

Topics:

- $GF(2)$
- Bits as field elements
- Polynomials over $GF(2)$
- Extension fields
- Why $GF(2^8)$ has 256 elements

Cryptography connection:

This prepares for AES multiplication and S-box construction.

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29. Polynomials with Binary Coefficients

Goal: Learn to read bit patterns as polynomials.

Topics:

- Byte-to-polynomial conversion
- Polynomial addition with exclusive OR
- Polynomial multiplication
- Cancellation of duplicate terms

Cryptography connection:

This directly supports finite field arithmetic.

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30. Irreducible Polynomials as Folding Rules

Goal: Understand irreducible polynomials without getting buried in abstraction.

Topics:

- Polynomial modular arithmetic
- Degree limits
- Reducing high powers
- Why irreducibility matters
- Prime-number analogy

Cryptography connection:

This is the heart of AES finite field arithmetic.

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31. Multiplication in $GF(2^8)$

Goal: Learn AES-style Galois field multiplication step by step.

Topics:

- Byte-to-polynomial conversion
- Polynomial multiplication
- Exclusive OR cancellation
- Reduction by $x^8 + x^4 + x^3 + x + 1$
- `xtime`
- Why AES uses `0x1B`

Cryptography connection:

This is essential for understanding AES MixColumns and the finite-field arithmetic behind the AES S-box.

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32. Multiplicative Inverses in $GF(2^8)$

Goal: Learn how nonzero bytes can have inverses in the AES field.

Topics:

- Inverse meaning
- Field multiplication
- Extended Euclidean algorithm for polynomials
- Lookup-table approach
- Brute-force inverse table
- Why 0 has no inverse

Cryptography connection:

AES uses multiplicative inverses to build its S-box.

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33. Building the AES S-box

Goal: Understand the math behind the AES substitution box.

Topics:

- Multiplicative inverse in $GF(2^8)$
- Affine transformation
- Why AES avoids simple linearity
- Nonlinearity
- No fixed points, gently
- S-box lookup tables

Cryptography connection:

This is one of the most important modern symmetric cipher tutorials.

Phase 6: Linear Algebra for Cryptography

This phase is crucial for diffusion, matrices, linear layers, and AES MixColumns.

34. Vectors and Matrices as Data Transformers

Goal: Understand vectors and matrices visually.

Topics:

- Vectors as lists of numbers
- Matrices as transformation machines
- Matrix multiplication
- Identity matrix
- Inverse matrix

Cryptography connection:

Many ciphers use matrices to mix data.

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35. Linear Transformations

Goal: Understand what linear means.

Topics:

- Preserving addition
- Preserving scalar multiplication
- Why exclusive OR-based transformations can be linear
- Visual intuition
- Invertible linear maps

Cryptography connection:

Permutation layers and diffusion layers are often linear.

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36. Matrices over $GF(2)$

Goal: Learn matrix operations where entries are only 0 or 1.

Topics:

- Exclusive OR as addition
- AND-like multiplication
- Binary matrices
- Invertibility
- Gaussian elimination over $GF(2)$

Cryptography connection:

Useful for linear feedback shift registers, diffusion layers, and linear cryptanalysis.

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37. Matrices over $GF(2^8)$

Goal: Understand AES MixColumns-style math.

Topics:

- Bytes as field elements
- Matrix multiplication using $GF(2^8)$
- MixColumns
- Inverse MixColumns
- Diffusion

Cryptography connection:

This explains one of the central layers of AES.

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38. Diffusion and Branch Number

Goal: Understand how small input changes spread.

Topics:

- Diffusion
- Avalanche effect
- Active S-boxes
- Branch number
- Maximum distance separable matrices, gently

Cryptography connection:

This gives language for evaluating cipher mixing layers.

Phase 7: Symmetric Cipher Structures

This phase connects math concepts to real cipher architecture.

39. Substitution-Permutation Networks

Goal: Understand how modern block ciphers combine confusion and diffusion.

Topics:

- S-boxes
- Permutation layers
- Round keys
- Repeated rounds
- Confusion
- Diffusion
- Avalanche behavior

Cryptography connection:

AES is a substitution-permutation network.

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40. Feistel Networks

Goal: Understand how non-invertible round functions can create invertible ciphers.

Topics:

- Split blocks
- Round function
- Exclusive OR mixing
- Swap
- Encryption/decryption symmetry
- Why the round function does not need to be reversible

Cryptography connection:

DES and many other ciphers use Feistel structures.

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41. Key Schedules

Goal: Understand how master keys become round keys.

Topics:

- Key expansion
- Round constants
- Rotations

- Substitutions
- Diffusion in key schedules
- Weak keys
- Related-key concerns

Cryptography connection:

This connects directly to experimental SPN design and round-key generation.

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42. Stream Ciphers and Keystreams

Goal: Understand ciphers that generate a pseudorandom mask.

Topics:

- Keystream generation
- Exclusive OR encryption
- State
- Synchronization
- Nonce or initialization vector
- Key reuse danger

Cryptography connection:

This extends one-time pad ideas into practical cryptography.

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43. Linear Feedback Shift Registers

Goal: Understand simple state machines over $GF(2)$.

Topics:

- Shift registers
- Feedback taps
- Recurrence relations
- Period
- Linearity
- Why pure linear feedback shift registers are not secure alone

Cryptography connection:

Important for stream ciphers and pseudorandom generators.

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44. Nonlinear Feedback and Combining Functions

Goal: Understand how stream ciphers resist linear attacks.

Topics:

- Nonlinear filters
- Combining linear feedback shift registers
- Boolean functions
- Correlation immunity
- Algebraic degree

Cryptography connection:

This leads toward ciphers like Trivium.

Phase 8: Cryptanalysis Foundations

This phase teaches how ciphers are examined and attacked mathematically.

45. What Makes a Cipher Breakable?

Goal: Build a vocabulary of cryptanalytic failure.

Topics:

- Brute force
- Known plaintext
- Chosen plaintext
- Chosen ciphertext
- Distinguishers
- Key recovery
- Structural weakness

Cryptography connection:

This frames the entire security mindset.

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46. Linear Cryptanalysis

Goal: Understand how linear approximations leak information.

Topics:

- Linear expressions
- Bias
- Parity
- Approximation tables
- Piling-up lemma, gently
- Key-bit guessing

Cryptography connection:

Important for evaluating S-boxes and block cipher structure.

---

47. Differential Cryptanalysis

Goal: Understand how input differences create output differences.

Topics:

- Exclusive OR differences
- Differential trails
- Difference distribution tables
- Active S-boxes
- Probability of trails

Cryptography connection:

One of the most important attacks on block ciphers.

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48. S-box Evaluation

Goal: Learn how to judge substitution boxes mathematically.

Topics:

- Nonlinearity
- Differential uniformity
- Linear approximation table
- Difference distribution table
- Fixed points
- Balance
- Avalanche criterion
- Algebraic degree

Cryptography connection:

This is extremely useful for experimental cipher design.

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49. Avalanche Testing

Goal: Learn how to measure bit-flip propagation.

Topics:

- Single-bit input flips
- Output bit changes
- Average avalanche
- Strict avalanche criterion
- Bit independence criterion
- Limits of avalanche testing

Cryptography connection:

This directly supports testing experimental cipher designs.

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50. Distinguishers and Random Permutations

Goal: Understand what it means for a cipher to look random.

Topics:

- Random functions
- Random permutations
- Distinguishing advantage
- Statistical tests
- Ideal cipher model, gently

Cryptography connection:

This gives a more formal way to talk about whether a construction behaves like a secure cipher.

Phase 9: Number Theory for Public-Key Cryptography

This phase moves from symmetric ciphers into public-key systems.

51. Prime Numbers and Divisibility

Goal: Build number theory foundations.

Topics:

- Divisibility
- Prime numbers
- Composite numbers
- Factorization
- Greatest common divisor
- Coprime numbers

Cryptography connection:

RSA and Diffie-Hellman depend heavily on prime-based structures.

---

52. Modular Exponentiation

Goal: Understand repeated multiplication modulo n.

Topics:

- Powers modulo n
- Fast exponentiation
- Square-and-multiply
- Why exponentiation is easy one way

Cryptography connection:

RSA and Diffie-Hellman both use modular exponentiation.

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53. Fermat's Little Theorem

Goal: Understand a key theorem about primes and modular powers.

Topics:

- Powers modulo prime
- Cycles
- Inverses
- Theorem intuition
- Simple examples

Cryptography connection:

Used in RSA reasoning and modular inverse computation.

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54. Euler’s Totient Function

Goal: Count invertible residues modulo n.

Topics:

- $\varphi(n)$
- Coprime counting
- $\varphi(p) = p - 1$
- $\varphi(pq) = (p - 1)(q - 1)$
- Why RSA uses it

Cryptography connection:

Essential for understanding RSA.

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55. Euler’s Theorem

Goal: Generalize Fermat’s Little Theorem.

Topics:

- Modular powers
- Coprime bases
- Exponent cycles
- Connection to inverses

Cryptography connection:

This is one of the mathematical foundations of RSA.

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56. RSA from the Ground Up

Goal: Understand RSA key generation, encryption, and decryption.

Topics:

- Choosing primes

- Modulus $n = pq$
- Totient
- Public exponent
- Private exponent
- Encryption
- Decryption
- Why decryption works

Cryptography connection:

This is the classic public-key cryptography tutorial.

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57. Why Factoring Matters

Goal: Understand why RSA security depends on factoring difficulty.

Topics:

- Public modulus
- Hidden primes
- Factoring problem
- Relationship to private key
- Key size intuition

Cryptography connection:

This connects abstract number theory to practical security.

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58. The Chinese Remainder Theorem

Goal: Understand how modular systems can be split and recombined.

Topics:

- Simultaneous congruences
- Remainders modulo different numbers
- Recombination
- RSA optimization

Cryptography connection:

Used in efficient RSA implementations.

Phase 10: Discrete Logarithms and Key Exchange

59. Cyclic Groups

Goal: Understand repeated multiplication as movement around a cycle.

Topics:

- Groups
- Generators
- Cyclic groups
- Order of an element
- Primitive roots, gently

Cryptography connection:

Diffie-Hellman lives in cyclic groups.

---

60. The Discrete Logarithm Problem

Goal: Understand why some exponentiation is easy forward and hard backward.

Topics:

- Modular exponentiation
- Generator powers
- Discrete logarithms
- One-way behavior
- Small examples

Cryptography connection:

This is the core hardness assumption behind Diffie-Hellman and related systems.

---

61. Diffie-Hellman Key Exchange

Goal: Understand how two people create a shared secret over an insecure channel.

Topics:

- Public parameters
- Private exponents
- Public shares

- Shared secret
- Man-in-the-middle risk

Cryptography connection:

This is the conceptual gateway into modern secure communication.

---

62. ElGamal Encryption and Signatures

Goal: Extend Diffie-Hellman ideas into encryption and signatures.

Topics:

- Ephemeral randomness
- Public keys
- Encryption
- Decryption
- Signatures

Cryptography connection:

ElGamal helps bridge Diffie-Hellman and digital signatures.

Phase 11: Elliptic Curve Cryptography

This is more advanced, but can be introduced gently if the group concept is already understood.

63. Elliptic Curves as Algebraic Shapes

Goal: Understand the basic equation and point set.

Topics:

- Curve equation
- Points on a curve
- Finite-field curves
- Point at infinity
- Visual intuition

Typical equation:

$$y^2 = x^3 + ax + b$$

Cryptography connection:

Elliptic curve cryptography uses points as group elements.

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64. Point Addition on Elliptic Curves

Goal: Understand the group operation.

Topics:

- Geometric addition over real numbers
- Finite-field point addition
- Doubling
- Inverses
- Identity element

Cryptography connection:

Point addition is the engine of elliptic curve cryptography.

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65. Scalar Multiplication

Goal: Understand repeated point addition.

Topics:

- kP
- Double-and-add
- Public key generation
- One-way behavior

Cryptography connection:

This is the elliptic curve version of modular exponentiation.

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66. Elliptic Curve Diffie-Hellman

Goal: Understand key exchange using elliptic curves.

Topics:

- Base point
- Private scalar
- Public point

- Shared secret
- Why the elliptic curve discrete logarithm problem matters

Cryptography connection:

This is a major foundation of modern Transport Layer Security.

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67. Digital Signatures: RSA, DSA, and ECDSA

Goal: Understand authentication and integrity mathematically.

Topics:

- Signing versus encryption
- Hash then sign
- Private signing key
- Public verification key
- Signature equations, gently

Cryptography connection:

Essential for certificates, software signing, secure boot, and authentication.

Phase 12: Hash Functions and Modern Primitives

68. What Hash Functions Are

Goal: Understand compression, fixed-length output, and one-way behavior.

Topics:

- Arbitrary input length
- Fixed output length
- Preimage resistance
- Second-preimage resistance
- Collision resistance

Cryptography connection:

Hashes are everywhere: passwords, signatures, certificates, integrity checks, blockchains, and message authentication.

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69. The Birthday Paradox

Goal: Understand why collisions happen faster than intuition expects.

Topics:

- Probability of shared birthdays
- Collision search
- Square-root effect
- Hash output size

Cryptography connection:

This explains why a 128-bit hash collision strength is roughly 64 bits.

---

70. Merkle-Damgård Construction

Goal: Understand one classic way to build hash functions.

Topics:

- Compression functions
- Chaining state
- Padding
- Length extension weakness

Cryptography connection:

Applies conceptually to older hash families such as MD5, SHA-1, and SHA-2.

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71. Sponge Construction

Goal: Understand SHA-3-style hashing.

Topics:

- Internal state
- Absorbing
- Squeezing
- Rate and capacity
- Permutation-based design

Cryptography connection:

SHA-3 and related modern designs use sponge construction.

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72. Message Authentication Codes

Goal: Understand keyed integrity.

Topics:

- Message authentication codes
- HMAC
- Authentication
- Integrity
- Why hash alone is not enough

Cryptography connection:

Message authentication codes are central to secure communication.

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73. Authenticated Encryption

Goal: Understand why confidentiality and integrity must be combined.

Topics:

- Encryption-only danger
- Encrypt-then-MAC
- Authenticated encryption with associated data
- Nonce requirements
- AES-GCM overview

Cryptography connection:

This is how modern systems actually protect messages.

Phase 13: Lattices and Post-Quantum Cryptography

This is advanced and should come after modular arithmetic, linear algebra, probability, and basic public-key cryptography.

74. Vectors, Length, and Distance

Goal: Build geometric intuition for lattices.

Topics:

- Vectors
- Norms
- Distance
- Dot product
- Nearest point intuition

Cryptography connection:

Lattice cryptography is geometry plus modular arithmetic plus probability.

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75. What Is a Lattice?

Goal: Understand lattices as repeating grids in higher dimensions.

Topics:

- Basis vectors
- Integer combinations
- Lattice points
- Good and bad bases
- Shortest vector problem

Cryptography connection:

Post-quantum cryptography often relies on lattice problems.

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76. Modular Linear Equations with Noise

Goal: Understand the Learning With Errors idea.

Topics:

- Linear equations
- Modular arithmetic
- Noise
- Why noise makes solving hard
- Small examples

Cryptography connection:

Learning With Errors is foundational for many post-quantum systems.

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77. Ring Learning With Errors

Goal: Understand how polynomial rings make lattice crypto efficient.

Topics:

- Polynomial arithmetic
- Modular polynomial rings
- Noise polynomials
- Structured lattices

Cryptography connection:

This relates to practical post-quantum systems such as Kyber-style designs.

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78. Post-Quantum Cryptography Overview

Goal: Understand why quantum computers change the landscape.

Topics:

- Shor’s algorithm impact
- Grover’s algorithm impact
- RSA and elliptic curve vulnerability
- Symmetric key size impact
- Lattice-based cryptography
- Hash-based signatures

Cryptography connection:

This is the modern frontier.

Suggested Course Organization

If this were shaped into a full personal course, the tutorials could be organized into the following units.

Unit 1: Mathematical Reading and Bit Fluency

1. How Mathematicians Use Symbols
2. Functions as Machines
3. Binary, Hexadecimal, and Bases
4. Bitwise Operations as Algebra

Unit 2: Modular Arithmetic and Classical Ciphers

1. Modular Arithmetic
2. Caesar Cipher
3. Multiplicative Inverses Modulo n
4. Euclidean Algorithm
5. Extended Euclidean Algorithm
6. Affine Cipher
7. Frequency Analysis
8. Vigenère Cipher
9. One-Time Pad

Unit 3: Permutations and Classical Machines

1. Permutations
2. Counting Keys
3. Transposition Ciphers
4. Rotor Machines
5. Polyalphabetic Systems

Unit 4: Probability, Entropy, and Randomness

1. Basic Probability
2. Random Variables
3. Entropy
4. Perfect Secrecy
5. Randomness Testing

Unit 5: Boolean Algebra and Finite Fields

1. Boolean Algebra
2. Exclusive OR as Addition Modulo 2
3. Boolean Functions
4. Algebraic Normal Form
5. From $GF(2)$ to $GF(2^n)$
6. Binary Polynomials
7. Irreducible Polynomials
8. Multiplication in $GF(2^8)$
9. Inverses in $GF(2^8)$

10. AES S-box

Unit 6: Linear Algebra and Symmetric Cipher Design

1. Vectors and Matrices
2. Linear Transformations
3. Matrices over $GF(2)$
4. Matrices over $GF(2^8)$
5. Diffusion and Branch Number
6. Substitution-Permutation Networks
7. Feistel Networks
8. Key Schedules
9. Stream Ciphers
10. Linear Feedback Shift Registers
11. Nonlinear Feedback

Unit 7: Cryptanalysis

1. What Makes a Cipher Breakable
2. Linear Cryptanalysis
3. Differential Cryptanalysis
4. S-box Evaluation
5. Avalanche Testing
6. Distinguishers and Random Permutations

Unit 8: Number Theory and Public-Key Cryptography

1. Prime Numbers
2. Modular Exponentiation
3. Fermat's Little Theorem
4. Euler's Totient Function
5. Euler's Theorem
6. RSA
7. Factoring
8. Chinese Remainder Theorem
9. Cyclic Groups
10. Discrete Logarithms
11. Diffie-Hellman
12. ElGamal

Unit 9: Elliptic Curve Cryptography

1. Elliptic Curves
2. Point Addition
3. Scalar Multiplication
4. Elliptic Curve Diffie-Hellman
5. Digital Signatures

Unit 10: Hashes, Authentication, and Modern Constructions

1. Hash Functions
2. Birthday Paradox
3. Merkle-Damgård Construction
4. Sponge Construction
5. Message Authentication Codes
6. Authenticated Encryption

Unit 11: Post-Quantum Foundations

1. Vectors, Length, and Distance
2. Lattices
3. Learning With Errors
4. Ring Learning With Errors
5. Post-Quantum Cryptography Overview

The Shorter Essential Path First

A full 78-tutorial path is valuable as a map, but it may be too much as a starting route. The following 20 tutorials form a practical first spine.

First Spine: Classical and Symmetric Foundations

1. Binary, Hexadecimal, and Bases
2. Bitwise Operations as Algebra
3. Functions as Machines
4. Modular Arithmetic
5. Multiplicative Inverses Modulo n
6. Euclidean Algorithm
7. Extended Euclidean Algorithm
8. Caesar and Affine Ciphers
9. Frequency Analysis
10. One-Time Pad
11. Permutations
12. Probability Basics
13. Entropy
14. Boolean Algebra
15. Exclusive OR as Addition Modulo 2
16. Binary Polynomials
17. Irreducible Polynomials
18. Multiplication in $GF(2^8)$
19. Inverses in $GF(2^8)$
20. AES S-box

This sequence gives an excellent foundation for understanding both classical ciphers and the guts of modern symmetric cryptography.

Second Spine: Symmetric Cipher Design and Analysis

1. Vectors and Matrices
2. Matrices over $GF(2)$
3. Matrices over $GF(2^8)$
4. Substitution-Permutation Networks
5. Feistel Networks
6. Key Schedules
7. Linear Feedback Shift Registers
8. Differential Cryptanalysis
9. Linear Cryptanalysis
10. S-box Evaluation

Third Spine: Public-Key Cryptography

1. Prime Numbers
2. Modular Exponentiation
3. Fermat's Little Theorem
4. Euler's Totient Function
5. Euler's Theorem
6. RSA
7. Cyclic Groups
8. Discrete Logarithms
9. Diffie-Hellman
10. Elliptic Curve Cryptography

Recommended Starting Order

For a personal quest into the mathematics of cryptography, the best starting sequence is probably:

1. Binary, Hexadecimal, and Bases
2. Bitwise Operations as Algebra
3. Functions as Machines
4. Modular Arithmetic
5. Multiplicative Inverses Modulo n
6. The Euclidean Algorithm
7. The Extended Euclidean Algorithm
8. Bytes as Polynomials
9. Irreducible Polynomials as Folding Rules

10. Multiplication in $GF(2^8)$
11. Inverses in $GF(2^8)$
12. The AES S-box

This route gives a tight bridge from ordinary numbers to AES mathematics without requiring a full expedition through every branch of mathematics first.

Guiding Principle

The curriculum should move from concrete to abstract:

1. Start with bits, bytes, and visible operations.
2. Move to modular arithmetic and classical ciphers.
3. Introduce invertibility, functions, and permutations.
4. Build toward finite fields and AES.
5. Then expand into linear algebra, cryptanalysis, public-key systems, and post-quantum topics.

In short:

> Build intuition first, formalism second, abstraction third.

That order keeps the mathematics from becoming a wall of symbols and turns it instead into a set of tools for understanding cryptographic machines.